

WEEK_5

SKILL

```
shivani@LAPTOP-NOJKE600:~$ echo 'Hello World'
Hello World
shivani@LAPTOP-NOJKE600:~$ echo 'Hello      World'
Hello      World
shivani@LAPTOP-NOJKE600:~$ name="Shivani"
shivani@LAPTOP-NOJKE600:~$ echo $name
Shivani
shivani@LAPTOP-NOJKE600:~$ echo "Hello $name"
Hello Shivani
shivani@LAPTOP-NOJKE600:~$ message='Hello $name'
shivani@LAPTOP-NOJKE600:~$ echo "$message"
Hello $name
shivani@LAPTOP-NOJKE600:~$ name="Shivani"
shivani@LAPTOP-NOJKE600:~$ printf '%s\n' '$name'
$name
shivani@LAPTOP-NOJKE600:~$ printf '%s\n' "$name"
Shivani
shivani@LAPTOP-NOJKE600:~$ echo '$HOME'
$HOME
shivani@LAPTOP-NOJKE600:~$ echo '$(date)'
$(date)
shivani@LAPTOP-NOJKE600:~$ echo '`date`'
`date`
```

To Apply Single Quotes, Preserve Literal Content, Ignore

Variable Expansion, Store Quoted Strings, Validate Parsing

Results, Test Edge Cases

Observation: Single quotes in the Linux shell preserve the literal content of a string. Variable expansion, command substitution, and interpretation of special characters are prevented inside single quotes. Quoted strings can be stored in shell variables and their parsing behavior can be verified using commands such as `echo` and `printf`. Edge cases such as empty strings, spaces, special characters, and single quotes within quoted strings were also tested.

```
shivani@LAPTOP-NOJKE600:~$ echo 'Hello World'
Hello World
shivani@LAPTOP-NOJKE600:~$ echo 'Hello    World'
Hello    World
shivani@LAPTOP-NOJKE600:~$ name="Shivani"
shivani@LAPTOP-NOJKE600:~$ echo $name
Shivani
shivani@LAPTOP-NOJKE600:~$ echo "Hello $name"
Hello Shivani
shivani@LAPTOP-NOJKE600:~$ message='Hello $name'
shivani@LAPTOP-NOJKE600:~$ echo "$message"
Hello $name
shivani@LAPTOP-NOJKE600:~$ name="Shivani"
shivani@LAPTOP-NOJKE600:~$ printf '%s\n' '$name'
$name
shivani@LAPTOP-NOJKE600:~$ printf '%s\n' "$name"
Shivani
shivani@LAPTOP-NOJKE600:~$ echo '$HOME'
$HOME
shivani@LAPTOP-NOJKE600:~$ echo '$(date)'
$(date)
shivani@LAPTOP-NOJKE600:~$ echo '`date`'
`date`
```

To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands

Observation: Double quotes were successfully used in the Linux shell to preserve spaces and treat multiple words as a single argument. Variable expansion was allowed inside double quotes, and command substitution using `$(command)` was also successfully performed. Nested tokens and escaped double quotes were tested to verify shell parsing. The outputs confirmed that double quotes preserve the structure of text while still allowing variable expansion and command substitution.